PROJECT SYFTIN

Unified System Specifications & Venture Architecture Core

Platform Concept: Distributed Zero-Trust Edge-Intelligence Network

Version Configuration: 2.0.0 (Overhauled Framework Architecture)

Date of Issue: July 2026

Operational Nodes: Native Multi-Platform Daemons (macOS Metal / Linux CUDA / Windows)

1. VISION BOARD

Corporate Strategy Manifesto

Project Syftin introduces a structural paradigm shift in high-throughput web data collection and extraction intelligence. Traditional data collection layers are facing systemic constraints: centralized cloud datacenter proxy nodes are universally flagged and blockaded by protective enterprise web perimeters (e.g., Cloudflare, Akamai), while the massive token-processing costs required to convert unformatted public HTML text into readable data objects via hosted model APIs (e.g., OpenAI, Anthropic) diminish platform margins.

Syftin reverses these vulnerabilities through a robust, pure Web2 decentralized model. We decouple operational task orchestration from actual structural computation loops. By deploying native background worker daemons across a distributed network of idle consumer computing setups, Syftin harnesses clean residential internet configurations alongside latent, distributed graphics card resources. The outcome is a hyper-secure data refinery network that bypasses traditional Cloudflare obstructions while driving operational processing expenses to near zero.

Strategic Alignment Grid

CORE NETWORK VISION	THE TARGETED STRATEGIC HURDLES
To construct the world's most resilient, highly compliant, decentralized edge-compute and data intelligence platform, operating entirely without Web3 cryptocurrency or token friction.	1. Absolute blockades of commercial data center IP ranges by anti-bot walls. 2. Cost-prohibitive cloud server overheads for LLM text processing. 3. Severe data leakage, packet manipulation, and operator falsification issues.

THE ENTERPRISE CUSTOMER ADVANTAGE	THE CONTRIBUTOR GROWTH ENGINE
Corporate buyers secure access to verified, highly accurate alternative datasets transformed into deterministic JSON structures at a 70% cost reduction, protected by automated edge PII sanitization.	Global consumer hardware hosts (e.g., college students) efficiently monetize their idle system VRAM and unmetered network infrastructure via a simple background daemon, earning cash paid directly via localized fintech rails.

Targeted Marketplace Segments

A. Revenue Channels (Data Customers)

- **AI Foundation Labs:** Engineering teams requiring localized, highly precise custom data segments to fine-tune specialized models and construct targeted vector stores.
- **Quantitative Analytics Groups:** Financial analysts requiring real-time alternative sentiment indicators and pricing streams for macro models.

- **E-Commerce Conglomerates:** Automated tracking managers requiring continuous product inventory updates without triggering anti-bot rate boundaries.

B. Network Infrastructure Pools (Contributors)

- **The Core Pilot Pool (College Students in India):** A dense concentration of tech-literate users possessing capable gaming hardware inside student residences equipped with unmetered institutional local network lines.
- **Global Homelab Operators:** Dedicated open-source server enthusiasts running multi-core systems looking to optimize network efficiency.

2. PRODUCT REQUIREMENTS DOCUMENT (PRD)

User Persona Journeys

The Micro-Hustle Contributor Experience

1. The contributor downloads a lightweight, cross-platform compiled desktop binary or Docker image (`Syftin Client Component`). The installer operates instantly with zero structural dependencies or Python environment requirements.
2. Upon initialization, the worker client runs a localized system profiling suite in the background to analyze device capabilities.
3. The client transitions to a quiet background monitoring state. Earnings accumulate in local currency (INR) and are paid out via immediate payment frameworks (UPI) once the user meets the baseline cash-out threshold ($\ge ₹500$).

The Corporate Enterprise Buyer Experience

1. The user logs into the Syftin Dashboard interface, defines target digital coordinates, and configures an expected structural JSON target layout scheme.
2. The user finances the operational task allocation via an embedded Web2 credit ledger system.
3. The user tracks processing performance through real-time telemetry metrics, downloading completed, cryptographically sealed datasets without data privacy liabilities.

Functional Requirements Matrix

ID	MODULE	FUNCTIONAL DEFINITION	CRITICALITY
FR-101	Telemetry Engine	Profile the host machine at launch to measure core metrics: CPU threads, available system RAM, and GPU/VRAM hardware constraints.	CRITICAL
FR-102	Dynamic Router	Poll and ingest orchestration tasks from the central cloud registry matching the specific host's capability tier.	CRITICAL
FR-103	Cross-Platform Core	Dynamically route internal model parsing drivers based on host OS layers (Metal on Apple Silicon, CUDA on Windows/Linux).	CRITICAL
FR-104	Edge PII Redactor	Scan output text buffers *prior* to encryption; scrub emails and phone configurations via localized regex pipelines.	HIGH
FR-105	Crypto Envelope	Apply asymmetrical AES-256 keys to seal extracted data inside volatile memory before transmitting payloads across the network.	CRITICAL

Non-Functional Constraints Matrix

- **Hardware Safety Shield:** Total continuous VRAM utilization on Tier-2 worker nodes must never exceed *M_{cap} = 3.0 ext{ GB}* to ensure safety margins on basic consumer GPUs (e.g., GTX 1650).
- **Zero Storage Fingerprint:** Node operators must never gain manual view access or interception avenues into target URLs or completed JSON structures. All processes are kept headless and hidden.
- **Metered Data Intercept:** The background daemon must pause network activities if host telemetry indicates a metered cell connection or active mobile hotspot tether to preserve mobile data caps.

3. TECHNICAL REQUIREMENTS DOCUMENT (TRD)

Cross-Platform Strategy & Tech Stack

To bypass local environment friction, the Syftin worker app must compile into a self-contained background daemon (written in Go or Rust). This binary interfaces directly with system drivers across macOS, Linux, and Windows, avoiding dependencies on local interpreters.

```
+-------------------------+ | Syftin Central Cloud | | (Supabase Core Cluster) |
+-------------------------+ ^ | Secure API Interface (TLS) v +-------------------------+ |
Syftin Worker Daemon | | (Rust/Go Compiled App Core)| +-------------------------+ |
+----------------------+----------------------+ | | v (macOS Platform Model) v (Linux
Platform Model) +-------------------------+ +-------------------------+ | Apple Silicon
Runtime | | Nvidia CUDA Runtime | | - Metal Framework (MPS) | | - CUDA Compute Pipelines | | -
Unified System Memory | | - Discrete VRAM Layouts | +-------------------------+
+-------------------------+
```

Operational Technology Stack

- **Target Workload Runtimes:** Asynchronous Go/Rust core worker engines.
- **Data Extraction Drivers:** Asynchronous `HTTPX` loops for raw HTML streaming; `Playwright Async` instances for client-side JavaScript execution.
- **Local LLM Middleware:** Embedded `Ollama` daemon integrations.
- **Target Model Configuration:** `qwen2.5:3b-instruct-q4_K_M` (4-bit quantization model, requiring ~2.0GB memory allocation).
- **State Coordination Cluster:** Supabase real-time cloud interfaces running over a PostgreSQL state engine.

Environment & Context Optimizations

Linux Driver Configuration Profile

```
# Restrict Ollama to single-threaded processing queues
export OLLAMA_NUM_PARALLEL=1
# Force computing pipelines strictly onto primary graphics slot
export CUDA_VISIBLE_DEVICES=0
```

macOS Driver Configuration Profile

```
# Restrict Ollama to single-threaded processing queues
export OLLAMA_NUM_PARALLEL=1
# Route model tensors directly through Apple Metal performance backends
export OLLAMA_GPU_DRIVER=metal
```

Network and Compliance Safety Controls

> **Hotspot Protection Guardrail:** The worker daemon must call local network gateway configuration indicators on a 60-second loop. If the active connection is flagged as a metered interface or Wi-Fi mobile tether, the engine must pause active task loops. Furthermore, all scraping activities must honor `robots.txt` declarations and implement a minimum 5-second rate-limiting delay per destination domain to eliminate DDoS risks.

4. ENGINEERING ARCHITECTURE SPECIFICATION

Automated System Spec Profiling Logic

When initialized, the Syftin daemon runs an automated benchmarking script. This reads available compute boundaries and drops the host into a specific task allocation slot:

ASSIGNED SLOT	MEASURED HARDWARE PROFILE	OPERATIONAL DUTY TYPE	REWARD SCALE
Tier 1: Scout	Integrated Graphics / CPU Only / < 8GB RAM	Pure HTTP Extraction, Network Proxy Routing	Base ($)
Tier 2: Ranger	4GB - 6GB VRAM (GTX 1650) / 8GB Mac M-Series	HTML Ingestion + 3B Local LLM Context Refining	Medium ($$)
Tier 3: Titan	≥ 12GB VRAM / ≥ 16GB Mac Unified Memory	Complex Multi-Step Agent Logic + 7B/14B Models	Premium ($$$)

End-to-End Zero-Trust Data Pipeline

To guarantee data correctness and prevent manipulation by node operators, the host system operates under a strict, isolated processing loop:

```
[Target Web Server Domain] | v (Download Stream) [Volatile System RAM Buffer Only] <--- (Never written to physical Hard Disk arrays) | v (Internal Local Transit) [Localized Core LLM Inference Parsing Engine (Single-Threaded Context)] | v (Clean JSON Object Extraction) [Edge Privacy Filters] <--- (Automatic structural Regex PII sanitization) | v (Raw JSON Object) [AES-256 Asymmetric Encryption] <--- (Sealed using Syftin Central Cloud Public Key) | v (Ciphertext In-Transit Payload) [Syftin Central Cloud Dashboard] <--- (Unreadable ciphertext; decrypted only at central hub)
```

Consensus Validation (Anti-Faking Mechanics)

To protect the platform from modified client apps generating fake or simulated data, high-value enterprise queries trigger a **Dual-Node Redundancy Constraint**. The central scheduler matches identical web retrieval operations to two independent worker nodes simultaneously. The incoming payloads are cryptographically compared. If an output mismatch occurs, the job is rerouted to a third node to break the tie, and the cheating node's metrics are permanently lowered.

5. BUSINESS REQUIREMENTS DOCUMENT (BRD)

The Unit Economic Disruption Advantage

Traditional centralized data intelligence competitors suffer from compressed margins because their processing pipelines run on expensive centralized cloud server instances, backed by high API costs from third-party model providers. Syftin shifts this entire computing footprint onto distributed edge devices, transforming consumer electricity and hardware capacities into high-efficiency data processing infrastructure. Platform operational costs remain virtually flat as processing volumes scale, allowing Syftin to undercut legacy market pricing by 70% while capturing premium gross margins.

The "Work Hard, Earn Hard" Operational Ledger

Syftin implements a precise compensation structure that allocates rewards based on the processing complexity and physical hardware load of completed tasks:

```
+-------------------------------------+ | Enterprise Client Task Deposit ($) |
+-------------------------------------+ | v +-------------------------------------+ |
Syftin 30% Platform Fee | +-------------------------------------+ | v (70% Direct Contributor
Outflow) ------------------------------------------------- | | | v v v [Tier 1: Scout]
[Tier 2: Ranger] [Tier 3: Titan] Pure Network Scraping 3B Model Data Refining 7B/14B Deep Agent
Refining (Base Payout) (Medium Payout) (Premium Payout)
```

Global Privacy and Data Compliance Architecture

To give enterprise buyers complete confidence, the Syftin edge engine applies automated formatting rules before data leaves the contributor node:

- **Automated Edge PII Scrubbing:** Payloads are parsed by a strict pattern-matching filter that flags personal identification indicators prior to encryption:

  ext{Pattern}_{ ext{Email}} ightarrow [a-zA-Z0-9._\%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]\{2,4\}

  Flagged fields are instantly rewritten into anonymous system tags (`[REDACTED_IDENTITY_MARKER]`). Private corporate text blocks never enter our cloud storage nodes or cross network transit layers in plain text.
- **Legitimate Interest Public Scope:** The platform restricts data extraction to publicly accessible interfaces that do not require credential verification or authentication bypasses, aligning with global data safety frameworks (GDPR and CCPA).

6. VENTURE-SCALE GROWTH ROADMAP

Chronological Implementation Milestones

Phase 1: Self-Validation Loop (Months 1–2)

- Deploy the unified Go/Rust scraping core prototype across your personal Mac and Linux laptops.
- Verify stability parameters for the database synchronization links connecting local nodes to a testing Supabase environment.
- Secure 3 alternative data extraction clients manually via freelance marketplaces (Upwork/Fiverr) to validate data output schemas and generate early revenue.

Phase 2: The Campus Pilot Expansion (Months 3–6)

- Wrap the verified edge client inside a lightweight background application wrapper optimized for retail distribution.
- Onboard an initial pilot group of 200 undergraduate college students in India, utilizing available student gaming setups connected to unmetered hostel Wi-Fi networks.
- Integrate immediate local settlement rails via **UPI payout APIs** (Razorpay/Cashfree), triggering automatic payouts straight to students' UPI IDs once they hit a ₹500 balance. This eliminates international banking friction and accelerates user acquisition.

Phase 3: General Public Launch & Venture Capital Sourcing (Months 7+)

- Open general public registration for the downloadable client app across Windows, macOS, and Linux platforms globally.
- Present the operational metrics of the distributed edge network to specialized early-stage deep-tech Venture Capital funds.
- Target an institutional **Seed Capital Raise of $1.5M to $2.5M**, using the capital to build advanced dataset validation engines and turn Syftin into the leading edge-intelligence marketplace for real-time web alternative data.